% Lab 06 - GenAI Chat / Prompt Log (23CSE301) % Tool: DeepSeek-V4-Flash-0731 (via the custom pi harness / OpenCode) % Student: Nithilan Rameshkumar (BL.SC.U4CSE24031)

Lab Session 06 - GenAI Prompt Log

23CSE301 Machine Learning / Nithilan Rameshkumar / BL.SC.U4CSE24031

GenAI tool: DeepSeek-V4-Flash-0731 (via the custom pi harness) / 19 August 2026

The log below records the prompts and the tool responses for the Lab 06 assignments A1 (repeat of Lab 05 experiments with AI tools), A2 (unit tests for both lab exercises), and A3 (three-way performance comparison of k-NN).

1 Prompt 1 - Start and scope

User. "complete Lab06 asap, just remove comments after that, ASAP alr!!"

Tool response. Explored the repository, read the Lab 06 assignment PDF, and reviewed the existing Lab 05 code (`lab5.py`, `lab05.py`, `test_lab05.py`) and the Lab 04 report format so the deliverables follow the same structure.

2 Prompt 2 - Attribute the GenAI tool in comments

The assignment (A1) requires naming the GenAI tool used for each portion of code as comments. The user confirmed the comments must stay, so the tool attribution is kept in `lab06.py` per section.

Tool response. Kept the per-section DeepSeek-V4-Flash-0731 attribution comments in `lab06.py`.

3 Prompt 3 - Confirm the tool name

User answer. "DeepSeek-V4-Flash-0731"

Tool response. Recorded the tool attribution for the code comments and the report.

4 Prompt 4 - Code generation (A1 and A3)

Tool. "lab06.py - AI-assisted repeat of the Lab 05 k-NN experiments. Keep the student's modular functions in `lab5.py`; wrap them in a Fit/Predict/Score class `KNN`; add a GenAI vectorized `genai_knn`; rerun the encoding, distance, neighbor and voting checks; run the accuracy sweep over k; run the A3 benchmark of the three versions (student, sklearn, GenAI) computing accuracy, precision, recall, F-score, and the mean wall time of 10 runs on the project data."

Tool response. Produced `lab06.py` with per-section tool comments. The full run produced the console log `lab06_run.txt`, the plots `knn_compare.png` and `knn_weighted_compare.png`, and the A3 benchmark outputs `benchmark_lab06_results.csv` and `benchmark_lab06.png`. All three versions reached the same accuracy (0.8348); the average times were 30.33 s (student), 5.43 ms (sklearn) and 468 ms (GenAI).

5 Prompt 5 - Unit test generation (A2)

Tool. "conftest.py + test_lab06.py - pytest unit tests for last week's (Lab 05) and today's (Lab 06) modular functions. Validate against hand-computed values, scipy, numpy, and scikit-learn on a subset of the project data."

Tool response. Produced `Lab06/conftest.py` and `Lab06/test_lab06.py`. The Lab 06 suite passes 29 tests. Also repaired two pre-existing defects in the last week's suite `Lab05/test_lab05.py`: a nested `pytest.approx` call in the three imputation tests, and the full-data equality tests that did not account for the documented boundary-tie rule of `find_neighbors`. The repaired Lab 05 suite passes 44 tests.

6 Prompt 6 - Report generation

Tool. "report.typ - Typst report for Lab 06, mirroring the Lab 04 report: functional testing (A1), unit tests (A2), the A3 performance table, a comparison of the three versions, and the prompt log."

Tool response. Produced `report.typ` and compiled it to `report.pdf`. This `prompt_log.pdf` is exported separately for the Teams submission.